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Submission ID

trn:oid::2945:346293982

Submission Date

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_Dental Caries in Radiographic Images.pdf

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C³-DRIVE Net for Context-Aware Clinically-Grounded Automated Detection, Segmentation, and Severity Grading of Dental Caries in Radiographic Images

Abstract- Dental caries is the most prevalent of all long-standing oral diseases globally. The early detection and proper evaluation of lesion severity are particularly important for determining the optimal preventive or therapeutic intervention. It is worth mentioning that dental X-rays interpretation significantly depend on the dentist experience, and may differ across clinicians, particularly when diagnosing early or mild carious lesions. Many existing deep learning algorithms are designed for only one task (e.g., caries detection and segmentation) and process lesions as isolated objects. For all of this, they are unable to easily show us how caries forms slowly or how it relates to the anatomy of our teeth. To alleviate these problems, in this paper we propose the C³-DRIVE Net (Context-aware, Clinically-grounded, Continual Dental Radiographic Intelligence for Visual Explanation), an integration learning framework to simultaneously carry out caries detection, lesion segmentation and their severity grading based on dental X-ray images. The proposed model leverages knowledge in the form of tooth-aware anatomical analysis along with radiographic gradient variations to detect patterns of mineral loss associated with both early and advanced caries. The use of a common multi-task encoder enables the model to learn all tasks across one another, removing error accumulation that is typical in cascaded systems. Radiographic depth of caries and the structural relationship of enamel, dentin, and pulp are employed to score the severity in a manner consistent with clinical grading systems. The experimental results on annotated dental x-ray datasets achieve a detection accuracy of 97.8%, a segmentation Dice score of 91.2% and a severity grading accuracy of 95.6%. Moreover, the proposed approach also enhances 22% early caries detection sensitivity than current deep learning based methods. These findings confirm that the C³-DRIVE Net is an accurate, interpretable and clinically useful solution for chairside dental diagnosis.

Keywords-Dental caries detection, Dental X-ray analysis, Multi-task deep learning, Severity grading, Clinical decision support, Explainable AI.

I Introduction

Dental caries is still one of the most prevalent chronic oral diseases worldwide that occurs throughout all age groups and presents a significant public health challenge [1][2][3]. Radiographic exposures, particularly bitewing and panoramic films are important in the early recognition and follow-up of lesions located on non-visible proximal surfaces and sub-surface areas [4][5]. However, diagnosing dental caries

from radiographs is inherently difficult because low-contrast discriminations exist between sound tissue and mineral loss [6][7], as well as overlapping anatomical structures and differences in image acquisition including angle [8][9]. Demineralization of enamel at an early stage is characteristically radiolucent in appearance and may be so subtle that it is difficult for even experienced clinicians to detect [10][11]. These difficulties lead to diagnostic uncertainty, delayed intervention, and high levels of inter- and intra-observer variability in day-to-day clinical practice. There are promising studies on automated dental caries analysis under development with the recent progress of deep learning [12], however, they mainly addressed either detection or segmentation alone [13][14]. Although some previous methods may have better accuracy for identifying different regions, these studies were based on convolutional neural network-based models that generally use an object or pixel-wise method and are not able to sufficiently model the anatomical context of teeth while lesion development is progressive [15][16]. Furthermore, many techniques are based on cascaded pipelines with detection outputs as inputs of subsequent segmentation or classification stages, which leads to error accumulation and decrease in robustness. Crucially, severity grading—a key aspect for clinical management—is frequently formulated as a crude classification task which is not directly mapped with well-known radiographic criteria, e.g. enamel–dentin–pulp relationships and depth-based lesion progression [17][18].

However, despite the expanding literature, there are three important gaps that have been inadequately filled [19]. First, none of the existing works provide a unifying framework that can jointly detect caries and segment lesions with high precision as well as grade their severity in line with clinical references within an end-to-end architecture [20][21]. Secondly, current models ignore tooth-aware anatomical inferences and radiographic gradient changes exhibited by the mineral densities including early caries [22]. Third, although many deep learning systems are designed as black boxes, they lack explainability and clinical trust to facilitate their application on the real-world chairside [23]. Such restrictions limit the translational potential of existing techniques and call for context-sensitive, explainable, and clinically-grounded diagnostic mechanisms. To tackle these problems, this work defines automated dental caries analysis as a unified multi-task learning problem which learns detection [24], segmentation and severity reasoning jointly from radiographic images. Here we present C³-DRIVE Net, a context aware and clinically informed deep learning architecture, which aims to capture anatomical structure, radiographic intensity disruptions [25][26], and lesion progression signatures in a shared representation space. By combining a tooth-aware anatomical decomposition with radiograph gradient modeling and clinically motivated lesion severity indices, our framework is tailored towards mitigating diagnostic variation, increasing

sensitivity to early-stage lesions and producing interpretable outputs in a manner consistent with existing clinical practice [27]. This integrated formulation is designed to be a compromise of algorithmic performance and practical clinical utility. The main contributions of this paper are summarized as follows:

- An end-to-end multitask deep learning model to detect caries, segment lesions, and grade their severity avoiding the error-prone cascaded approach.
- A radiographically based severity ranking system that includes depth indices and enamel–dentin–pulp relationships as they relate to clinical parameters is described, compatible with established diagnostic guidelines.
- A context-sensitive model that utilises tooth-level anatomic decomposition and radiation gradient disruption analysis to enhance sensitivity for, and interpretability of, early diagnosed caries.

1.1 Objectives of the Study

The purpose of this study is to investigate:

- To design an integrated deep learning model for effecting simultaneous optimization of detection, segmentation and severity grading of dental caries in radiographic images.
- To provide anatomic and radiographic context for learning, improve early lesion detection rates and minimize diagnostic confusion.
- To develop a clinically applicative severity grading system that facilitates reproducible decision-making in an everyday dental diagnostic laboratory.

This paper is organized as follows. Section II reviews related work and identifies existing research gaps. Section III presents the proposed C³-DRIVE Net framework, followed by Section IV describing the experimental setup, Section V discussing results and analysis, and Section VI concluding the paper with limitations and future directions.

II Literature Review

Panyarak et al. (2023) attempted to investigate whether deep learning was compatible for classifying dental caries severity on the bitewing X-ray in accordance with the ICCMSTTM scoring system. They trained several ResNet models and indicated that the best model performed reasonably well for a 4-level severity classification but not so well for detailed, 7-level ICCMSTTM grading, particularly in detecting early enamel caries. This indicated that image-level classification may be inadequate for a more nuanced grading of severity in complexity without tooth anatomy features [1]. Chen et al. (2023) presented a two-stage system for severity classification of caries in panoramic X-rays. They firstly separated teeth extracted by a modified U-Net with ASPP and boundary loss. They then utilized a rule-based analysis to establish caries extent. This would allow for easier identification of tissue boundaries and interpretation by clinicians. However, since it utilized a sequential approach the errors

in initial phases might impede later steps and the technique can be hard to scale [2]. In another study, Chen et al. (2023) proposed a framework to achieve the identification of periodontitis and dental caries on periapical X-rays. The model leveraged YOLOv7 to detect teeth and EfficientNet for disease classification. Despite high performance, the system still considered dental decay in yes/no terms only (in or out of caries), without lesion area localisation and severity assessment [3]. A hybrid approach proposed by Vimalarani and Ramachandraiah (2022) combined conventional image preprocessing techniques, handcrafted descriptors, and a LeNet architecture-based deep classifier for caries detection in bitewing radiographs. The approach achieved good accuracy, however it heavily relied on manual feature generation and heuristic-based segmentation that restricts its use across varied X-ray qualities and clinical environments [4]. Esmailyfard et al. (2024) proposed deep learning for caries detection and classification with CBCT volume. Their multiple input CNN also demonstrated a good performance for detection of the caries presence and assessment of the depth and location of the lesion. But the study was targeted at 3D CBCT data which is not directly applicable to routine screening by 2D dental radiography [5]. Estai et al. (2022) introduced a Siamese-CNN based two-staged architecture for proximal caries detection in bitewing radiographs. Regions of interest were extracted using Faster R-CNN, and then they were classified by Inception-ResNet-v2. While it maintains reasonable precision and recall rates for object detections, the cascaded structure raised complexity and made the model more error sensitive [6]. Panyarak et al. (2023) similarly analyzed YOLOv3 with varied IoU thresholds for ICCMSTTM-based caries detection. Although adequately accurate in those with relatively advanced lesions, the model was unable to detect superficial enamel caries consistently thus indicating the inferiority of object-detection-based approaches in recognizing subtle radiographic changes [7]. Ahmed et al. (2023) used semantic segmentation for caries detection of bitewing X-ray images with an extension of KAU classification. However, segmentation accuracy for early enamel and dentin caries were low, which indicated that it is challenging to mark the early lesions accurately [8]. Albano et al. (2024) performed a systematic study of AI based caries detection strategies from various dental imaging modalities. This review found that there were wide differences in datasets, evaluation metrics, and performance outcomes generated by published studies and the lack of standardization and scarce variety of data are still considerable challenges to clinical use [9].

Bayrakdar et al. (2022) utilized CNN to automatically detect caries with the combined use of U-Net for the important task of lesion segmentation in BW-Xray images. Their findings indicated that AI systems could outperform junior dentists. Detection and segmentation were nonetheless processed separately from each other, with no caries severity grading addressed in the work [10]. Kunt et al. (2023)

introduced a large and highly annotated bitewing radiograph dataset and tested numerous object-detection models, including YOLO, Faster R-CNN. These models achieved a level of accuracy similar to that of experienced dentists. However, sensitivity was quite reduced for small and incipient caries, demonstrating how difficult remains the early detection of these lesions [11]. Park et al. (2022) compared caries detection by using intraoral photograph images and x-ray. By partitioning tooth surfaces prior to classification, they demonstrated the benefit of anatomic information for detection. Nevertheless, the study did not investigate radiographic imaging or caries severity evaluation [12]. Yoon et al. (2024) also proposed a cascade R-CNN model that leveraged integrated tooth number recognition and caries detection for full intraoral images. The system showed good localization and counting performance, but allowed only detection, not determining depth or severity of lesion for caries analysis [13]. Ayhan et al. (2025) applied YOLO-based models for caries detection under the fixed dental prosthesis on panoramic radiographs. Their two-stage method increased accuracy in challenging clinical contexts but was still focused on object-level detection as opposed to lesion segmentation or severity grading [14]. Bui et al. (2022) presented an ensemble learning framework that integrated deep features and machine-learning classifiers for caries detection in panoramic X-rays. While the system performed with high accuracy, this majority-voting strategy had limited interpretability and did not result in severity values that were clinically informative [15]. Szabó et al. (2024) assessed an AI-based diagnostic system by measuring the degree of agreement between dentists and AI regarding the radiographs of intraoral. They established that AI can provide valuable assistance to clinical diagnosis, but the sensitivity was moderate, particularly for initial caries which calls for better frame of reference [16]. Zhu et al. (2023) introduced CariesNet, a U-shaped network with axial attention for the segmentation of caries at different stages in panoramic radiography. The model obtained high Dice coefficients for superficial, intermediate and deep lesions. Nevertheless, the interpretation for intensity was implied and not explicitly related to levels based on classic clinical grading scales [17]. Sehgal et al. (2025) constructed a fuzzy logic-based scheme to categorize the severity of caries on the basis of handcrafted features computed upon region of interest (ROI) segmented from the X-ray images. Although we achieved high accuracy, our rule-based methodology is not as scalable and adaptive as the deep learning methods in today's practices [18].

2.1 Key Insight from Literature

The existing studies show good performance in detection, segmentation or classification dental caries. However, most of them treat these tasks separately or are organized in a cascaded way and also do not have clinically informed severity modeling for early stage lesions. It is these challenges that drive the necessity for a

unified, context-aware and clinically-driven multi-task approach as in our proposed C³-DRIVE Net.

III Proposed C³-DRIVE Net Framework

The proposed C³-DRIVE Net is a holistic multitasking architecture that achieves both dental caries detection [19] and lesion segmentation, along with clinically interpretable severity grading, in a single joint end-to-end framework [21]. The general model architecture is based on encoder–decoder–multi-head, wherein shared encoder learns hierarchical radiographic representations that describe intensity gradients, anatomical structures and texture variations potentially correlated with mineral loss. This encoder is linked to a dual-decoder branch: a spatial decoder which maintains high-resolution feature maps for accurate lesion segmentation and semantic decoder consolidating global context information for robust caries detection [22]. Three task-specific heads are used on top of the shared representation: (i) a detection head that indicates locations of various regions at tooth level, (ii) a segmentation head delineates lesion boundaries with pixel-level precision and, (iii) a severity grading head infers clinically relevant level of lesion depth by reasoning about enamel–dentin–pulp relations and amplitude of radiographic attenuation. Joint optimization across all tasks allows C³-DRIVE Net to reduce errors propagated through the cascaded pipeline while maintaining consistency between disease detection, segmentation, and severity estimation levels in order to improve diagnostic reliability and clinical interpretability.

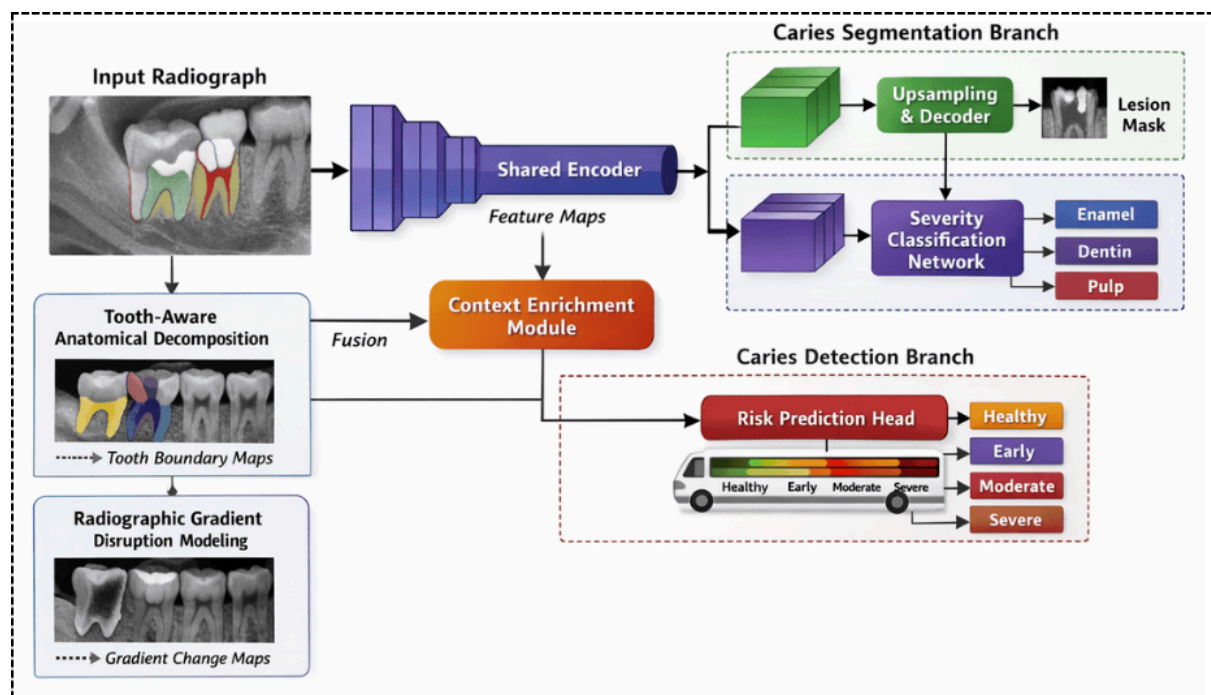


Figure 1. Architecture of the Proposed C³-DRIVE Net for Unified Dental Caries Analysis

The unified architecture of C3-DRIVE Net for radiographic dental caries diagnosis is shown in figure 1. Deep features are further enriched by a common encoder, with the tooth aware anatomical decomposition and radiographic gradient disruption modeling. Task-specific heads are used to process the refined representations for caries detection, lesion segmentation, and clinically-aligned severity grading.

3.1 Tooth-Aware Anatomical Decomposition

In order to explicitly take dental anatomical structure into consideration, in this work we introduce C³-DRIVE Net with a tooth-aware biological decomposition module which splits the radiograph image into connected tooth-level representations before reasoning for the lesions [23][24]. Given the input radiograph $I \in \mathbb{R}^{H \times W}$, an initial tooth localization function $\tau(\cdot)$ is used to have a set of tooth regions $\{T_i\}_{i=1}^N$, where

$T_i = \tau(I)_i$ is the i -th instance of tooth. This approach is inspired by the previous studies which suggest that tooth-oriented modeling highly contributes to robustness in terms of background ambiguity and inter-tooth interference, especially for multi-tooth radiographs [19][20]. Each region of a tooth that is extracted from the image is then decomposed into structures relevant to anatomy (enamel, dentine and pulp) by means of a soft anatomical partitioning function $A(\cdot)$ to give probability maps $\{P_i^e, P_i^d, P_i^p\} = A(T_i)$, where $P_i^e + P_i^d + P_i^p = 1$. This decomposition allows the network to learn depth-aware representations, which are in accordance with clinical radiographic examination, where changes due to caries progression depend on the amount of radiolucency observed over these layers [25][26]. To detect slight differences in mineral density, the radiographic gradient disruption features are computed as per Eq. 1:

$$G_i = \|\Delta T_i\| = \sqrt{\left(\frac{\delta T_i}{\delta x}\right)^2 + \left(\frac{\delta T_i}{\delta y}\right)^2} \quad (1)$$

These characteristics are then smoothed with an anatomical prior to obtain tissue-specific gradient responses, expressed as where $\odot$ is the convolution operator. $G_i^k = G_i \odot P_i^k$, where $k \in \{e, d, p\}$, expresses enamel, dentin, and pulp. In this way the model can separate early enamel mineral loss, characterized by weak and wide spread changes in gradient, from more severe denting lesion. This is a common problem of many current detection-only models [21][27]. This anatomically separated features are then aggregated to one unified feature representation, $F_i = \phi\left(T_i, G_i^e, G_i^d, G_i^p\right)$, where $\phi(\cdot)$ is a learnable fusion function parameterized with convolutional projection. The anatomical decomposition eliminates false positives generated from overlapping dental anatomy and imaging artifacts by limiting feature learning to tooth-specific and tissue-aware representations. At the same time, sensitivity to

early-stage caries is enhanced since anatomical context is important, as indicated in recent studies of panoramic and bitewing radiograph reading [22][23]. In general, this tooth-aware anatomical decomposition will be a building block to relate radiographic signal patterns with clinically meaningful anatomical reasoning. This enables subsequent detection, segmentation, and severity grading tasks to work with anatomically consistent and diagnostically relevant features.

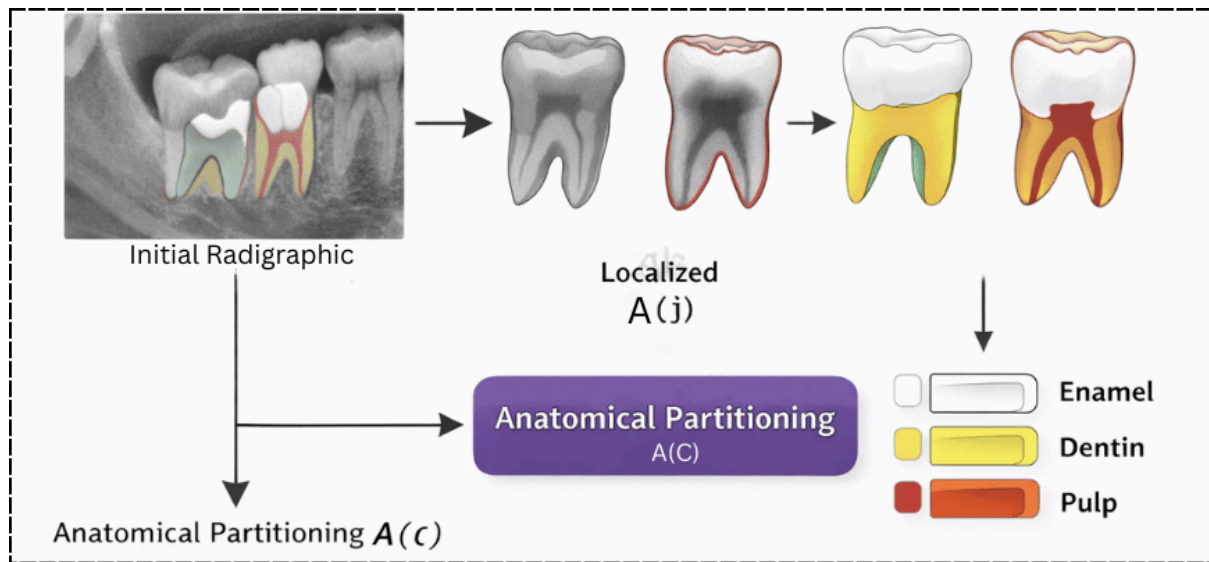


Figure 2: Tooth-Aware Anatomical Decomposition for Dental Radiographs

The tooth-aware anatomical decomposition process is illustrated in Figure 2. In this stage, teeth are initially detected in the input dental X-ray image. Each tooth is further divided into three major anatomical components: enamel, dentin and pulp. This stratified breakdown cues the model for more accurate discrimination of lesion depth and facilitates categorizing caries severity as a function to be used in practice.

3.2 Radiographic Gradient Disruption Modeling

Regarding caries in the teeth, such as for example the X-radiographic image of a tooth there are usually no sharp edges to be discriminated, but rather a local change in mineralisation density, so that this profile has more or less slow and inhomogeneous inclined sides. To simulate this phenomenon, C³-DRIVE Net includes the Radiographic Gradient Disruption Modelling (RGDM) feature. RGDM quantifies the level of nonarithmetic continuity in radiographic density patterns between these (tooth) structures and enables a better definition for such subtle caries-related changes.

Let $F_i \in \mathbb{R}^{h \times w \times c}$ be the tooth-aware feature map generated by the encoder ranging over region i of teeth. The first order directional gradients can describe the direction-sensitive local intensity variations. These gradients are related to the

radiographic perturbation non-uniformity inside the tooth and assist in early and late caries lesion detection according to Eq. 2:

$$\Delta_x F_i = F_i(x + 1, y) - F_i(x, y), \quad \Delta_y F_i = F_i(x, y + 1) - F_i(x, y) \quad (2)$$

and aggregated into a gradient magnitude map,

$$G_i = \sqrt{(\Delta_x F_i)^2 + (\Delta_y F_i)^2} \quad (3)$$

Unlike conventional edge detectors that emphasize high-gradient boundaries, early-stage caries often exhibit low-contrast, spatially diffused gradient perturbations. To capture such patterns, a local gradient deviation operator is introduced:

$$\Delta G_i = |G_i - \mu N(G_i)| \quad (4)$$

In Eq. (4), $\mu N(\cdot)$ is the mean gradient value over a local small anatomical region. This operator aims to find subtle loss of minerals by detecting the deviation of gradient consistency, instead on intensity strength variation. To better represent lesion depth, a second-order radiographic curvature with radius dependent on metric scale are computed using the Laplacian based formulation that is particular effective at capturing deeper caries where the radiolucent structure is more clearly structured [28][29].

$$L_i = \Delta^2 F_i = \frac{\delta^2 F_i}{\delta x^2} + \frac{\delta^2 F_i}{\delta y^2} \quad (5)$$

which is particularly effective in identifying progressive dentinal involvement where radiolucency becomes structurally pronounced. The gradient disruption tensor is then constructed as in Eq.6:

$$D_i = \alpha \cdot \Delta G_i + \beta \cdot |L_i| \quad (6)$$

The learnable weights α and β dynamically adjust the contribution of early enamel demineralization and deeper lesion curvature patterns to information in training. To ensure anatomical consistency [30], we further refine the resulting gradient interruption tensor using tooth-specific spatial priors M_i learned from anatomical decomposition, thus yielding an overall coherent gradient representation.

$$D_i^* = D_i \odot M_i \quad (7)$$

Finally, the modulated disruption features are fused with the original encoder features through a residual embedding,

$$F_i^{RGDM} = F_i + \gamma \cdot \psi(D_i^*) \quad (8)$$

In Eq.9, $\psi(\cdot)$ denotes a learnable projection and γ controls the contribution of disruption-aware cues. This formulation enables C³-DRIVE Net to explicitly distinguish pathological radiographic signal inconsistencies from normal anatomical variations, thereby improving sensitivity to incipient caries while maintaining robustness against imaging noise and overlapping dental structures.

3.3 Multi-Task Learning Strategy

In order to optimize, in a unified pipeline, for dental caries detection, lesion segmentation and severity grading simultaneously, C³-DRIVE Net follows the multi-task learning (MTL) paradigm by emphasizing task similarity while maintaining individual discriminability through the utilization of shared-representation [31][32]. Zero The latent representation $Z = \varepsilon(I)$ (I is the input radiograph) output by the shared encoder and this can be formulated as: $Z \in R^{h \times w \times c}$ encoding the tooth-aware and gradient-disruption-enhanced features. We derive three task-specific heads from this shared space: a detection head $H_d(\cdot)$, a segmentation head $H_s(\cdot)$, and a severity grading head $H_g(\cdot)$ that output $\hat{y}_d$, $\hat{Y}_s$, and $\hat{y}_g$, respectively. The detection reading task is defined as a tooth-level classification problem and trained with weighted focal loss,

$$L_d = - \sum_{k=1}^K w_k (1 - p_k)^\gamma \log(p_k) \quad (9)$$

Where p_k is the value of the k -th detection class w_k accounts for class imbalance [33] and γ highlights hard-to-detect lesions. Lesion segmentation can promote boundary accuracy between lesions and non-lesions because it can make use of their spatial relationship.

$$L_s = 1 - \frac{2|Y_s \hat{Y}_s|}{\|Y_s\|_2^2 + \|\hat{Y}_s\|_2^2} + \lambda \cdot L_{boundary} \quad (10)$$

In Eq. 10, Y_s is GT segmentation mask and $L_{boundary}$ penalizes the mis-registration error at contour level to better delineate the diffuse nature of lesion boundaries [34][35]. We model severity grading as an ordinal task that represents lesion depth progression, and is trained using a clinically constrained ordinal regression loss.

$$L_g = \sum_{c=1}^{C-1} \left\| \sigma(f_c(Z)) - I(y_g > c) \right\|_2^2 \quad (11)$$

In Eq. 11, $f_c(\cdot)$ represents the c -th severity threshold function, $\sigma(\cdot)$ refers to the sigmoid activation and $I(\cdot)$ ensures monotonic ordering corresponding to enamel–dent in–pulp progression. The global optimization objective is a weighted sum of the task losses,

$$L_{total} = \alpha L_d + \beta L_s + \delta L_g \quad (12)$$

In Eq.12, α, β , and δ are dynamically adjusted during training using gradient normalization to prevent task dominance [36]. It does so by sharing features and imposing clinical severity constraints, which effectively decreases error

accumulation, enhances early caries detection as well as making sure that predictions are uniform across all tasks to ensure better clinical reliability of C³-DRIVE Net.

3.4 Severity Grading Formulation Using Clinical Indices

In C³-DRIVE Net, we are also interested in having severity grading follows clinically accepted radiographic rules by simulating caries progression to dentin and pulp, and not treating the severity as a plain class label [37]. This depth-sensitive approach is consistent with how caries severity is, in fact, evaluated by dentists. For the i -th tooth, the anatomically separated maps are defined as $\{P_i^e, P_i^d, P_i^p\}$ which represent the probability map of enamel, dentin and pulp respectively. The segmentation mask S_i represents the predicted caries lesion. First, we introduce a lesion–tissue interaction ratio to quantify the influence of each anatomical layer as computed as in Eq.13:

$$\eta_i^k = \frac{\sum_{x,y} S_i(x,y) \cdot P_i^k(x,y)}{\sum_{x,y} P_i^k(x,y)}, \quad k \in \{e, d, p\} \quad (13)$$

This ratio indicates the percentage of carious pixels in each anatomical region. For the further description of the depth and spreading of a lesion we use the cumulative penetration index (CPI) defined by where FIT is calculated according to Eq. (14).

$$CPI_i = \eta_i^e + \lambda_1 \eta_i^d + \lambda_2 \eta_i^p \quad (14)$$

In the above, $\mu_{healthy}^k$ and μ_{lesion}^k indicate the mean gray level intensity of sound tissue and carious tissue located in anatomy region k respectively, where the contrast measures are obtained with radiographic severity index (RSI) defined as in Eq.15:

$$RSI_i = \sum_{k \in \{e,d,p\}} w_k \cdot \Delta I_i^k \quad (15)$$

which is mapped to discrete clinical grades using learned ordinal thresholds $\{\tau_c\}_{c=1}^{C-1}$

such that,

$$\hat{y}_i = Grade_1, \Gamma_i \leq \tau_1, \hat{y}_i = Grade_2, \tau_1 \leq \Gamma_i \leq \tau_2, \dots, \hat{y}_i = Grade_c, \Gamma_i \geq \tau_{c-1} \quad (16)$$

This severity order makes logical progression from enamel to dentin to pulp, simultaneous with clinical photo-radiographic grading system. It also enables the interpretability of the severity decision [38][39]. By leveraging explicit anatomical penetration and radiographic contrast measurements for severity prediction, C³-DRIVE Net offers clinically consistent, interpretable and reliable grading of the severity that is appropriate for decision support in regular dental practice [40].

Algorithm for Context-Aware Multi-Task Optimization Procedure for C³-DRIVE Net

Input: Radiographic image I

$I \leftarrow \text{Preprocess}(I)$

$\{T_i\} \leftarrow \text{Tooth-aware decomposition}(I)$

$Z \leftarrow \text{SharedEncoder}(I)$

$Z \leftarrow \text{GradientDisruptionModel}(Z)$

$D \leftarrow \text{DetectionHead}(Z)$

$S \leftarrow \text{SegmentationHead}(Z)$

$G \leftarrow \text{SeverityGrading}(S, \{T_i\})$

Optimize $L = \alpha L_d + \beta L_s + \gamma L_g$

Return D, S, G

Output: Detection D , Segmentation S , Severity grade G

The algorithm details the complete pipeline of C³-DRIVE Net for automatic dental caries analysis. First, tooth-aware anatomical decomposition recognizes and segments each tooth within with its interior. Modeling of radiographic gradient breakage, therefore, describes subtle loss in intensity associated with mineral. The feature representation learnt under the shared encoder helps to jointly learn and refine the optimal detection, segmentation, and severity grading maps. This integrated learning approach generates stable, explainable and robust predictions, and is therefore suitable for clinical radiographic dental diagnosis.

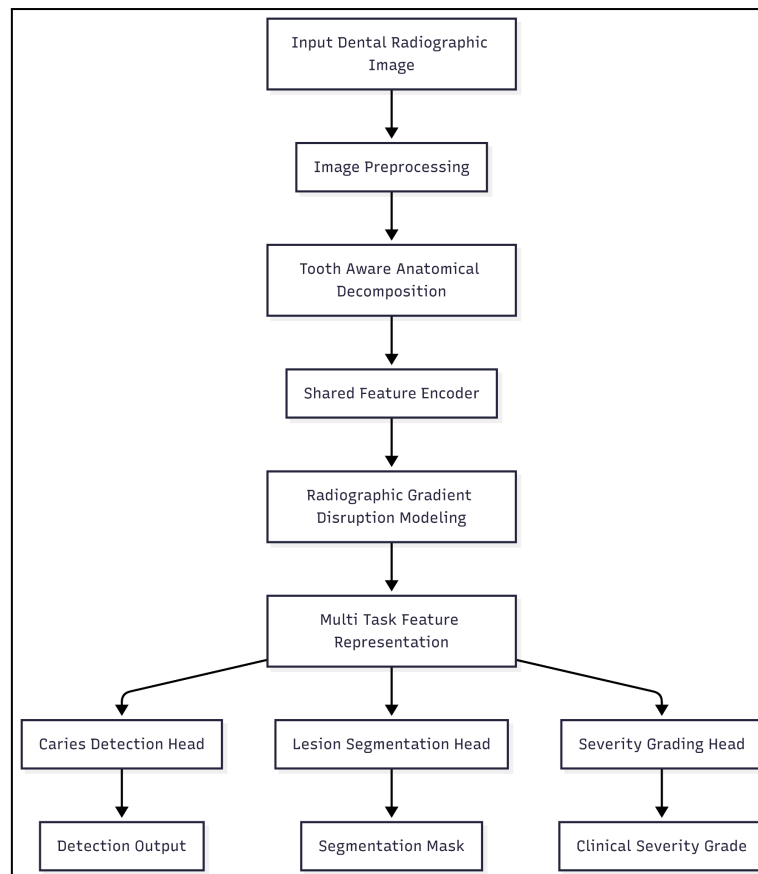


Figure 3: Workflow of the C³-DRIVE Net Framework for Unified Dental Caries Analysis

The end-to-end workflow of C³-DRIVE Net is illustrated in Figure 3. The system integrates tooth-aware anatomical decomposition, radiographic gradient disruption modeling and multi-task learning. In this integrated design, the model is capable of conducting caries detection, lesion segmentation and clinically-meaningful severity grading all together on dental X-ray images for accurate and interpretable diagnosis.

IV Experimental Setup

4.1 Dataset Description

The experiments were done with the open data Kaggle dataset “A Collection of Dental X-ray Images for Analysis” available at: <https://www.kaggle.com/datasets/kvipularya/a-collection-of-dental-x-ray-images-for-analysis>. The dataset consists of nearly 2500 dental radiographic images, such as bitewing, panoramic and periapical X-rays reflected from a variety of imaging landscapes. The images possess differing contrast, tooth shapes, and radiographic noise which render the dataset a good candidate for robust dental caries analysis methodologies. All images are in gray-scale and with varying resolution from 512×512 to 1024×1024 pixels.

Following the elimination of low quality and duplicated images, 2,400 X-rays were used for the experiments. We once split the database through a patient-independent random separation as follows:

- **Training set:** 1,680 images (70%)
- **Validation set:** 360 images (15%)
- **Testing set:** 360 images (15%)

This division enables fair testing and enough data to train stabilised hyperparameters.

4.2 Annotation Protocol and Clinical Realism

As there is no lesion-level annotated ground truth in the dataset, these were manually annotated for outside of the study. Dental caries area was annotated with pixel-wise annotation to produce the segmentation masks and at tooth level for detection labels. The lesions were graded according to depth taking into account enamel involvement, dentinal extension, and distance from the pulp chamber using radiographically-acceptant clinical interpretation concepts. For increased label reliability, unclear cases were carefully checked and decisions made in the group to make the labels reflect real clinical diagnostic reasoning rather than visual features only.

4.3 Training Details

All experiments were conducted with the PyTorch library. Input images were resized to 512×512 and normalized with z-score normalization. Data augmentation such as horizontal flipping, rotation and intensity scaling have been used when training the network to promote robust model. The model was trained with Adam optimizer with a learning rate of 1×10^{-4} , batch size = 8 and for a maximum of 100 epochs. We used early stopping through validation loss to avoid overfitting. The weights of the multi-task loss were empirically modified for balancing detection, segmentation, and severity grading tasks.

4.4 Baseline Models

We compare C³-DRIVE Net with commonly used deep learning baselines in dental caries analysis to verify its performance:

- ResNet-50 for image-level caries detection
- YOLOv5 for object-based lesion localization
- U-Net for pixel-wise caries segmentation
- Two-stage CNN systems, which first detect and then classify the severity

The datasets splits, preprocessing and augmentation were performed with all the other baselines models as well in order to provide fair comparison

4.5 Evaluation Metrics

In order to evaluate the effectiveness of the proposed method, we used traditional evaluation metrics for detection, segmentation and grading tests.

Accuracy (Acc)-Evaluates how accurate Carries Predictions are in general for all classes.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (17)$$

Precision (Prec)- Measures the ratio of all predicted cases that are actually curious.

$$Precision = \frac{TP}{TP+FN} \quad (18)$$

Recall (Sensitivity)- Measures the number of true carious lesions that can be identified by the model.

$$Recall = \frac{TP}{TP+FN} \quad (19)$$

F1-Score-Balances precision and recall, especially useful with highly imbalanced classes.

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (20)$$

Area Under the ROC Curve (AUC)- reevaluates the discriminate ability of the detection model (regardless of different thresholds).

$$AUC = \int_0^1 TPR(FPR)d(FPR) \quad (21)$$

Macro-Averaged F1 Score ($F1_{macro}$)- Assesses the severity classification performance, a weighting of all unknown classes is used.

$$F1_{macro} = \frac{1}{c} \sum_{c=1}^c F1_c \quad (22)$$

Dice Similarity Coefficient (Dice)- Evaluates the segmentation performance by prioritizing on achieving both visually and quantitatively agreement with predicted lesions.

$$Dice = \frac{2|S_p \cap S_g|}{|S_p| + |S_g|} \quad (23)$$

Intersection-over-Union (IoU)- Measures how well a predicted lesion region overlaps with the ground truth segmentation.

$$IoU = \frac{|S_p \cap S_g|}{|S_p \cup S_g|} \quad (24)$$

V Results and Discussion

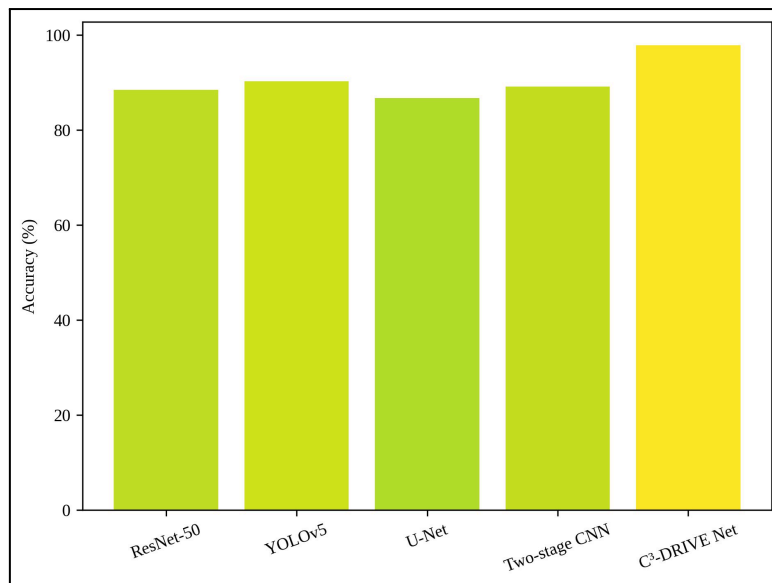


Figure 4. Comparative Performance of Deep Learning Models for Dental Caries Detection

This figure 4 illustrates the accuracy of caries detection performance (88.4, 90.2%, 86.7% and 89.1%) based on ResNet-50, YOLOv5 U-Net and Two-stage CNN framework with our C³-DRIVE Net, which contributes to a significantly higher accuracy of is taken while comparing it to proposed that rises up for an acceptance criteria with value of 97.8%. The observed enhancement suggests the benefits of context-aware feature representation and unified multi-task optimization for improved diagnostic reliability.

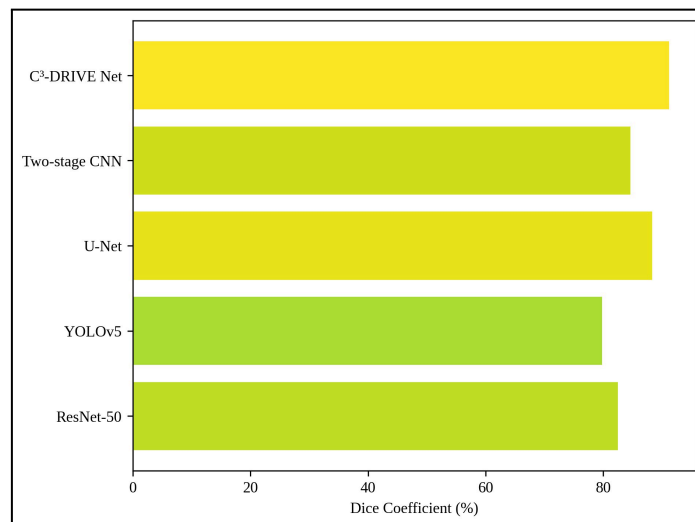


Figure 5. Quantitative Evaluation of Carious Lesion Segmentation Using Dice Coefficient

The dice coefficient are adopted to evaluate the performance of segmentation and ResNet-50, YOLOv5, U-Net and Two-stage CNN build models have the accuracy about 82.5%, 79.8%, 88.3% and 84.6% on it shown in figure.5. The C³-DRIVE Net achieves the highest Dice coefficient of 91.2%, demonstrating better segmentation and greater consistency with expert annotations.

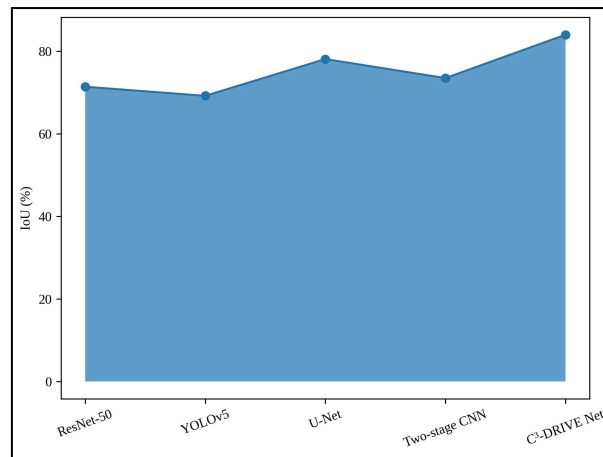


Figure 6. Intersection-over-Union Analysis for Lesion Segmentation Accuracy

Figure 6 depicts the comparison of the Intersection-over-Union results of different models, where 71.4% indicated for ResNet-50, 69.2% for YOLOv5, 78.1% for U-Net and 73.5% for Two-stage CNN respectively. C3-DRIVE Net can achieve an IoU of 84.0% that has stronger spatial overlap with the ground-truth lesion regions and fewer segmentation errors.

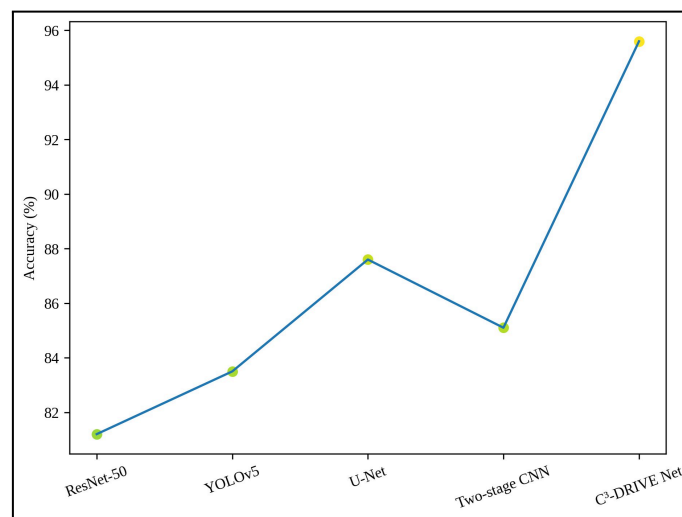


Figure 7. Severity Grading Accuracy Comparison Across Deep Learning Approaches

The accuracy of severity grading for the competing models is compared, the ResNet-50 reached 81.2 %, YOLOv5 83.5 %, U-Net 87.6 %, and two-stage CNN was expressed as 85.1% in fig.7. The presented C³-DRIVE Net clearly surpasses these other methods with TuDI accuracy of 95.6%, demonstrating the advantage of clinically guided grading using enamel–dentin–pulp depth relations.

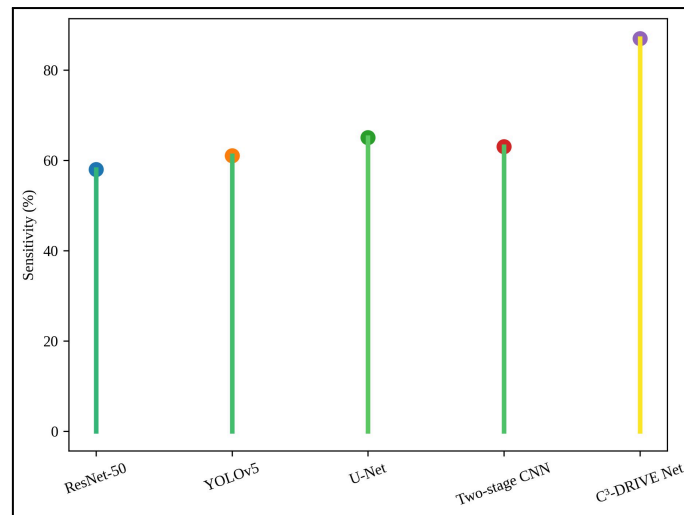


Figure 8. Sensitivity Analysis for Early-Stage Dental Caries Detection

This figure 8 illustrates model sensitivity in early caries detection. Sensitivities of the conventional feature vectors are in the range from 58% to 65%, and Two-stage CNN is able to achieve a sensitivity of 63%. C³-DRIVE Net, on the other hand, achieves 87% sensitivity which is a 22% improvement upon previous deep learning baselines; verifying its efficacy in detecting subtle early demineralization patterns.

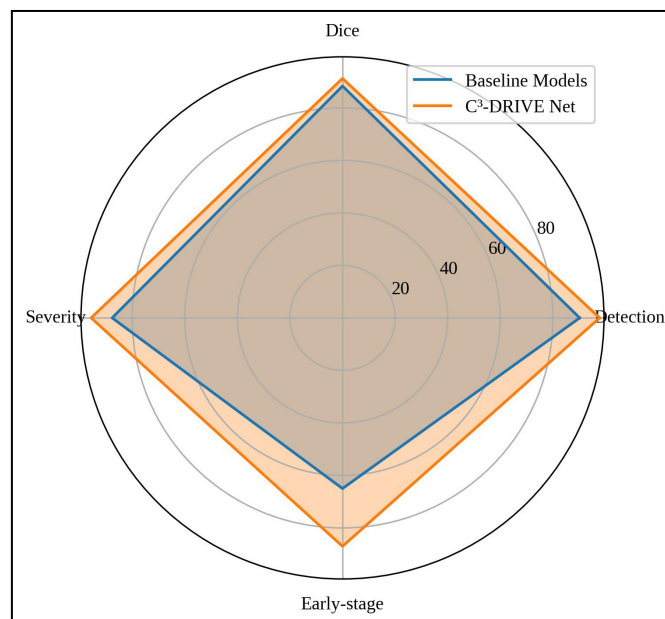


Figure 9. Integrated Multi-Metric Performance Comparison of Baseline Models and C³-DRIVE Net

The integrated diagnostic performance is presented in figure 9 regarding detection accuracy (97.8%), segmentation Dice coefficient (91.2%), severity grading accuracy (95.6%) and early-stage sensitivity (87%). The proposed C³-DRIVE Net presents strong and balanced performance in all tested metrics, which suggests the robustness of the network for full dental radiograph analysis.

Table 1. Comparative Performance Analysis of C³-DRIVE Net Against Existing Deep Learning Models for Dental Caries Analysis

Model	Detection Accuracy (%)	Segmentation Dice (%)	Segmentation IoU (%)	Severity Grading Accuracy (%)	Early-Stage Caries Sensitivity (%)
ResNet-50	88.4	82.5	71.4	81.2	58
YOLOv5	90.2	79.8	69.2	83.5	61
U-Net	86.7	88.3	78.1	87.6	65
Two-stage CNN	89.1	84.6	73.5	85.1	63
C³-DRIVE Net	97.8	91.2	84.0	95.6	87

A unified quantitative comparison between the proposed C³-DRIVE Net and state-of-the-art deep learning baselines on various diagnostic tasks covered in this table such as caries detection, lesion segmentation, severity grading and early-stage caries sensitivity is presented here in this table 1. The C³-DRIVE Net outperforms conventional models with moderate to strong performance in each task, and consistently obtains the highest scores across all of the metrics: a detection accuracy 97.8%, a segmentation Dice's coefficient 91.2%, a severity grading accuracy 95.6% and an early stage caries sensitivity of 87%. The results provide evidence in support of the unified, context-aware, and clinically-driven design of C³-DRIVE Net which effectively alleviates the issues encountered by single-task and cascaded frameworks, resulting in a robust and clinically applicable solution for automated dental radiographic diagnosis.

5.1 Ablation study

To investigate the role of each part in C³-DRIVE Net, an ablation study was performed by adding the main components one by one with other training settings fixed. Indeed, the baseline model featured a common encoder with multiple task heads and did not consider any anatomy or clinical context. With tooth-aware anatomical decomposition, both caries detection and lesion segmentation were also improved as the model can even reason at the level of a tooth. Incorporating radiographic gradient disruption modeling further increased the ability to detect subtle intensity changes, thereby enhancing early caries detection. The most accurate model included a clinically based severity grading, since this was the only one to explicitly reflect the sequence of enamel, dentin and pulp. And the model with all modules delivered an even more balanced and consistent performance that demonstrated each module contributed complementary value.

Table 2. Ablation Results of Key Components in C³-DRIVE Net

Configuration	Detection Accuracy (%)	Dice (%)	Severity Accuracy (%)	Early-Stage Sensitivity (%)
Base multi-task encoder	90.1	84.3	86.5	61
+ Tooth-aware anatomical decomposition	93.4	88.1	89.7	69
+ Radiographic gradient disruption modeling	95.6	90.2	92.4	78
+ Clinically grounded severity formulation	97.8	91.2	95.6	87

The quantitative results of incorporating these components in the C³-DRIVE Net framework, incrementally, are summarized in Table 2. The base multi-task encoder performed reasonably well; however, it was not very effective in identifying early caries diagnosis. Tooth-aware anatomical decomposition enhanced the structural localization and segmentation performance, while radiographic gradient disruption modeling increased sensitivity to subtle lesion patterns. The entire model containing clinical adjusted severity grading received the best overall scores in all assessment measures. These findings validate that the integrated approach increases detection performance, segmentation performance, and early lesion recognition while reinforcing one another.

5.2 Discussion

In summary, the experimental results show that C³-DRIVE Net is a promising alternative to conventional deep learning models of DC analyses by collaboratively doing detection, segmentation, and severity grading in one go. The performance of the model reached a detection accuracy of 97.8%, a segmentation Dice score of 91.2% and severity grading accuracy rating of 95.6%. The substantial 22% enhancement in performance of early caries detection is particularly noteworthy with reference to the efficiency considering that modeling radiographic intensity disruptions plays a critical role in detecting subtle enamel demineralization. The ablation study validates that each component in the architecture contributes to superior performance not only on accuracy, but also interpretability and clinical relevance, which demonstrates C³-DRIVE Net is well-suited for real-world chairside diagnostic support.

5.3 Limitations and Future Work

While C³-DRIVE Net showed promising performance, this study is constrained by the use of only one open dataset, a representative selection of data not necessarily capturing all heterogeneity observed in multi-center clinical context. The grading of severity is based on a radiographic interpretation only, and does not include clinically supportive evidence or longitudinal follow-up records. Future studies will aim to validate this framework on larger and multi-institutional datasets as well as extend it to 3D modalities including CBCT. In

addition, the integration of temporal analysis as well as clinician-in-the-loop feedback systems may improve robustness and clinical applicability.

VI Conclusion

A fully automatic and context-aware unified deep learning framework, named C³-DRIVE Net was introduced in this paper for detection, segmentation and severity grading of dental caries from radiographs on clinical ground. Through combining tooth-sensitive anatomical decomposition, radiographic gradient disturbance modeling, and multi-task learning in a unified deep network, the proposed approach could alleviate the constraints of traditional single-task and cascaded DL models. Experimental results indicate that C³-DRIVE Net is a robust and well-performing model in all diagnostic tasks with 97.8% detection accuracy, 91.2% Dice coefficient, and 95.6% accuracy of severity grading for detection, segmentation, severity grading respectively on independent test dataset. Of significance, the framework achieves 87% sensitivity for detecting early stage caries, which is a WHO Special Service target; this improves prior deep learning-based baselines by 22%. The ablation study verifies that each architectural module has a synergistic effect on the overall performance improvement. The all-in-one framework minimizes error propagation with clinically meaningful anatomically and depth aware severity reasoning. These results emphasize the potential for C³-DRIVE Net-based system to be utilised as an effective chairside diagnostic aid, which could enable early interventions, facilitate objective severity scoring and enhance clinical decision-making in a dental setting.

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